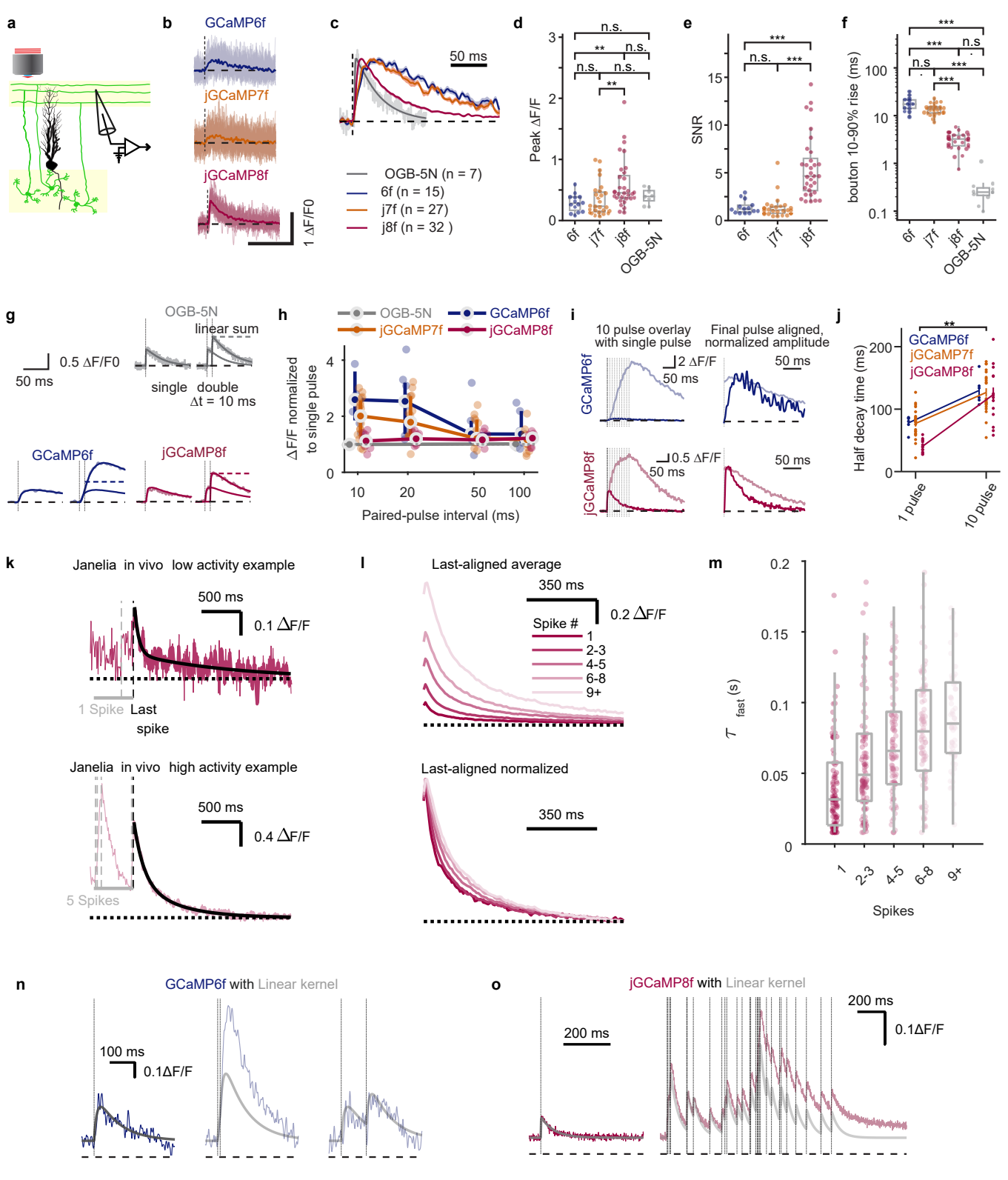

a
b
GCaMP6f
jGCaMP7f
jGCaMP8f
1 ΔF/F0
c
50 ms
OGB-5N (n = 7)
6f (n = 15)
j7f (n = 27)
j8f (n = 32)
d
Peak ΔF/F
n.s.
**
n.s.
n.s.
**
6f
j7f
j8f
OGB-5N
e
SNR
***
n.s.
***
f
bouton 10-90% rise (ms)
***
***
n.s
n.s
***
g
OGB-5N
linear sum
0.5 ΔF/F0
50 ms
single
double
Δt = 10 ms
GCaMP6f
jGCaMP8f
h
OGB-5N
GCaMP6f
jGCaMP7f
jGCaMP8f
ΔF/F normalized to single pulse
Paired-pulse interval (ms)
i
10 pulse overlay with single pulse
Final pulse aligned, normalized amplitude
2 ΔF/F
50 ms
0.5 ΔF/F
j
Half decay time (ms)
**
1 pulse
10 pulse
k
Janelia in vivo low activity example
500 ms
0.1 ΔF/F
1 Spike
Last spike
Janelia in vivo high activity example
0.4 ΔF/F
5 Spikes
l
Last-aligned average
350 ms
0.2 ΔF/F
Spike #
1
2-3
4-5
6-8
9+
Last-aligned normalized
m
τ fast (s)
Spikes
n
GCaMP6f with Linear kernel
100 ms
0.1ΔF/F
o
jGCaMP8f with Linear kernel
200 ms
0.1ΔF/F